

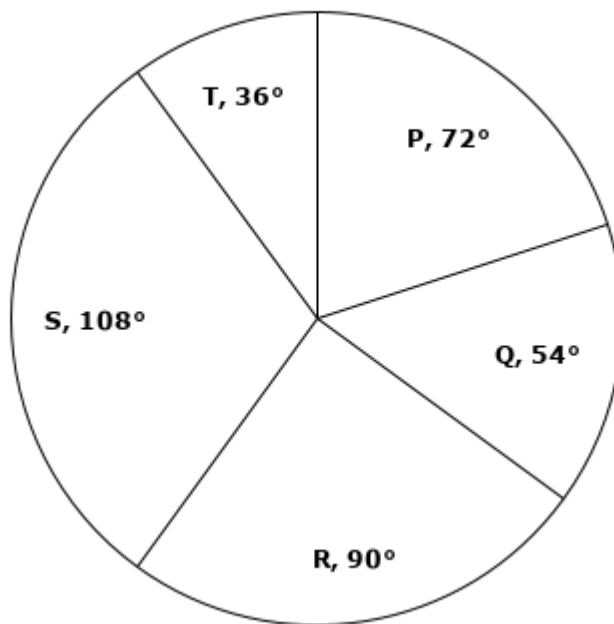
1. Questions

Study the following information carefully and answer the questions given below.

The pie chart data given below shows the degree distribution of the number of accounts opened by each branch, out of the total number of accounts opened by all the given branches together.

Note: The number of accounts opened by branch 'R' is 450.

The degree distribution of the number of accounts opened



If the number of accounts opened by Branch R is 90 less, and the number of accounts opened by Branch T is 60 more, then find the ratio of the number of accounts opened by Branch R to Branch T now.

- a. 4 : 3
- b. 3 : 2
- c. 5 : 4
- d. 2 : 1
- e. 5 : 3

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2. Questions

If the number of accounts opened by Branch T is 36 more, then the new number of accounts opened by Branch T is what percentage of the number of accounts opened by Branch P?

- a. 50%
- b. 75%
- c. 60%
- d. 40%

e. 80%

3. Questions

If the number of accounts opened by Branch U was 40% more than those opened by Branch Q and Branch T together, then find the number of accounts opened by Branch U.

- a. 630
- b. 540
- c. 720
- d. 450
- e. 270

4. Questions

If 25% and 20% of the accounts opened by Branch P and Branch S, respectively, are zero-balance accounts. Find the total number of zero-balance accounts opened by Branches P and S together.

- a. 182
- b. 198
- c. 212
- d. 204
- e. 190

5. Questions

Find the difference between the number of accounts opened by Branch S and the number of accounts opened by Branch Q.

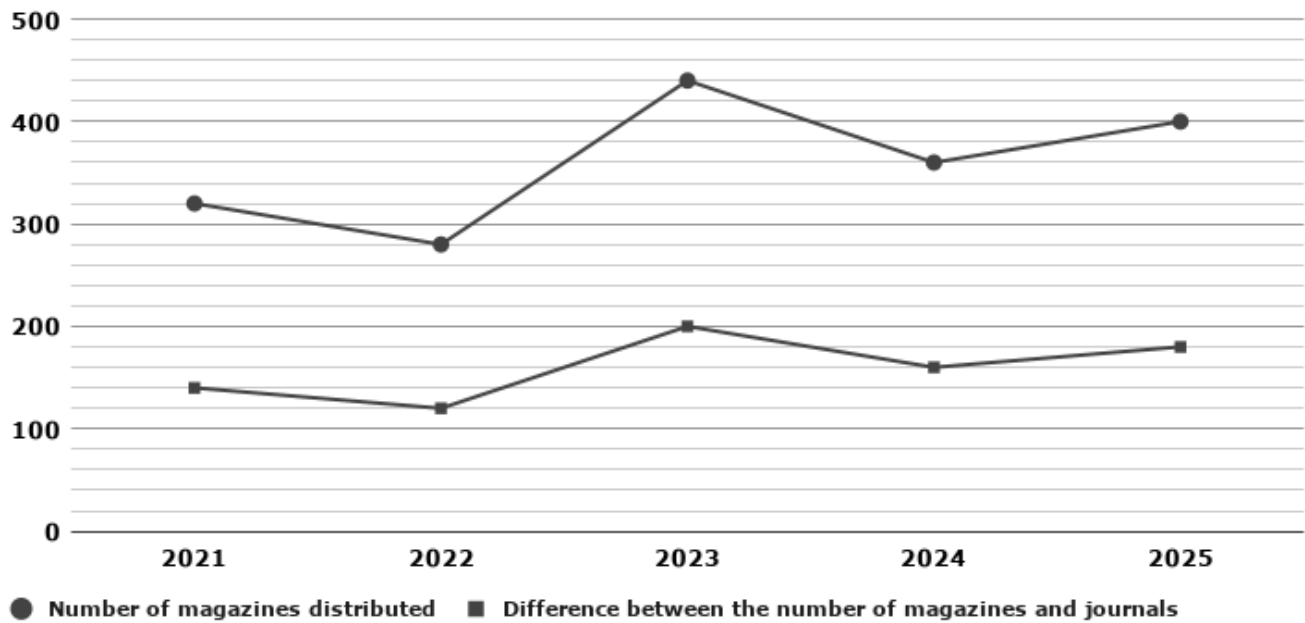
- a. 210
- b. 240
- c. 290
- d. 250
- e. 270

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6. Questions

Study the following information carefully and answer the questions given below.

The line graph below shows the total number of magazines distributed and the difference between the number of magazines and journals distributed by a publishing house across five different years—2021, 2022, 2023, 2024, and 2025.



Note: The number of magazines distributed by the publishing house is greater than the number of journals distributed in all the given years.

Find the difference between the total number of journals distributed in 2024 and the total number of magazines distributed in 2022.

- a. 90
- b. 80
- c. 60
- d. 70
- e. 120

7. Questions

If $\frac{1}{4}$ of the magazines distributed in 2022 are academic magazines and the remaining magazines are sports magazines, find the total number of sports magazines distributed in 2022.

- a. 70
- b. 140
- c. 280
- d. 35
- e. 210

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8. Questions

The total number of magazines and journals distributed in 2024 is what percentage more than the total number of magazines distributed in 2021?

- a. 75%
- b. 60%
- c. 80%
- d. 65%
- e. 70%

9. Questions

Find the average number of journals distributed by the publishing house in 2021, 2024, and 2025.

- a. 210
- b. 190
- c. 200
- d. 180
- e. 220

10. Questions

If the total number of magazines distributed in 2020 was 20% more than that in 2022, then find the total number of magazines distributed in 2020.

- a. 336
- b. 216
- c. 276
- d. 324
- e. 384

11. Questions

A boat covers 60 km downstream in 4 hours. If the speed of the boat in still water is 4 times the speed of the stream, find the time taken by the boat to cover 36 km upstream.

- a. 2 hours
- b. 3 hours
- c. 4 hours
- d. 5 hours
- e. 6 hours

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12. Questions

The ratio of the ages of Dev and Divya 4 years ago was 3:4, respectively. If after 8 years from now,

the ratio of the ages of Dev and Divya becomes 5:6, then find the present age of Dev.

- a. 24 years
- b. 28 years
- c. 22 years
- d. 26 years
- e. 30 years

13. Questions

A sold a watch to B at a 20% profit, and B sold it to C at a 10% loss for Rs. 1080. Find the cost price of the watch for A.

- a. Rs. 800
- b. Rs. 900
- c. Rs. 1000
- d. Rs. 1200
- e. Rs. 1500

14. Questions

Rice of variety 'A' costing Rs. 80/kg and rice of variety 'B' costing Rs. 50/kg are mixed in the ratio of 3:2 respectively. Find the cost price of the resulting mixture per kg.

- a. Rs. 62
- b. Rs. 65
- c. Rs. 68
- d. Rs. 70
- e. Rs. 72

15. Questions

P and Q started a business together such that the initial investment of Q is Rs. 15000. After six months, Q left the business, and P increased his investment by Rs. 3000. At the end of the year, the ratio of the profit share of P to Q is 4:3, then find the initial investment of P.

- a. Rs. 9500
- b. Rs. 7500
- c. Rs. 9000
- d. Rs. 8000
- e. Rs. 8500

16. Questions

The area of the square is 900 cm^2 . The ratio of the side of the square to the length of a rectangle is 5: 6, and the breadth of the rectangle is 15 cm. Find the perimeter of the rectangle.

- a. 108 cm
- b. 102 cm
- c. 96 cm
- d. 112 cm
- e. 120 cm

17. Questions

The simple interest earned on investing Rs. '450x at 15% per annum for 4 years is Rs. 5400. Find the LCM of $(x + 2)$ and $(x + 6)$.

- a. 252
- b. 312
- c. 286
- d. 143
- e. 114

18. Questions

Rohit spent 15% of his monthly income on rent, 10% of his income on travel, 20% of the remaining income on food, half of the remaining income on other expenses, and he saved the remaining income. If his savings are Rs. 9000, then find the amount spent on travel.

- a. Rs. 2500
- b. Rs. 3000
- c. Rs. 4500
- d. Rs. 3600
- e. Rs. 4000

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19. Questions

The speed of a car is 45 km/hr. If the car travels for 3 hours and 20 minutes to cover a certain distance, then find the total distance covered by the car.

- a. 160 km
- b. 120 km
- c. 135 km

d. 150 km

e. 180 km

20. Questions

The sum of three consecutive even numbers is 56 more than the average of the same three numbers. Find the product of the smallest and largest numbers among them.

a. 780

b. 680

c. 720

d. 880

e. 800

21. Questions

What value will come in place of the question mark (?) in the following question?

$$(18 * 12) + (15 * 16) - (14 * 10) = ?$$

a. 312

b. 316

c. 320

d. 324

e. 308

22. Questions

$$\sqrt{784} + \sqrt{169} = ? + 18 - \sqrt{256}$$

a. 35

b. 37

c. 39

d. 41

e. 43

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23. Questions

$$45\% * 600 + 30\% * 450 = ? + 105$$

a. 290

b. 280

c. 310

d. 320

e. 300

24. Questions

$$(5/8) * 720 - (4/9) * 630 = ? * 5$$

a. 32

b. 34

c. 36

d. 38

e. 40

25. Questions

$$16^2 - 14^2 + (180 / 4) = ? + 5^2$$

a. 80

b. 75

c. 70

d. 85

e. 90

26. Questions

What approximate value will come in place of the question mark (?) in the following question?

$$\sqrt[3]{1330.98} + 24.99\% \text{ of } 480.05 = ? / 4.01$$

a. 508

b. 580

c. 524

d. 428

e. 632

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27. Questions

$$(3.99)^4 * (16.01)^2 \div (63.99)^2 = 4^?$$

a. 1

b. 4

c. 3

d. 2

e. 5

28. Questions

$$1199.98 \div 23.99 + 44.98 * 12.01 = ?$$

a. 580

b. 590

c. 600

d. 610

e. 620

29. Questions

$$\sqrt{(? + 84.01)} = 15.99 * 4.01 - 25.01 * 1.99$$

a. 192

b. 94

c. 112

d. 136

e. 162

30. Questions

$$(74.99\% \text{ of } 640.01) \div [(1/3) * 179.98] = ?$$

a. 18

b. 8

c. 48

d. 32

e. 72

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31. Questions

What value will come in the place of (?) in the following number series?

11, 22, 44, ?, 176, 352

a. 88

b. 86

c. 90

d. 92

e. 94

32. Questions

47, ?, 64, 113, 77, 102

a. 124

b. 126

c. 130

d. 132

e. 128

33. Questions

?, 8, 8, 12, 24, 60

a. 14

b. 18

c. 20

d. 16

e. 32

34. Questions

212, 196, 178, 158, ?, 112

a. 132

b. 134

c. 136

d. 138

e. 140

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35. Questions

50, 55, 65, 80, 100, ?

a. 120

b. 125

c. 130

d. 135

e. 140

36. Questions

Find the wrong term in the given series.

15, 25, 39, 63, 95, 135

- a. 15
- b. 25
- c. 39
- d. 63
- e. 135

37. Questions

10, 14, 30, 64, 130, 230

- a. 30
- b. 14
- c. 230
- d. 64
- e. 130

38. Questions

100, 99, 90, 64, 0, -125

- a. 90
- b. 64
- c. 99
- d. 0
- e. -125

39. Questions

5, 13, 37, 109, 320, 973

- a. 13
- b. 37
- c. 320
- d. 109

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e. 973

40. Questions

50, 67, 86, 109, 138, 170

- a. 67
- b. 138
- c. 109
- d. 86
- e. 170

41. Questions

In each of the following questions, two equations are given. You have to solve both equations to find the relation between x and y.

I). $x^2 + 8x + 15 = 0$

II). $y^2 - 6y - 27 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x < y$
- d. $x \leq y$
- e. $x = y$ or a relationship cannot be established.

42. Questions

I). $x^2 + 19x + 84 = 0$

II). $y^2 + 15y + 56 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x < y$
- d. $x \leq y$
- e. $x = y$ or a relationship cannot be established.

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43. Questions

I). $x^2 - 14x + 48 = 0$

II). $y^2 - 6y + 8 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x < y$
- d. $x \leq y$
- e. $x = y$ or a relationship cannot be established.

44. Questions

I). $x^2 - 12x + 35 = 0$

II). $y^2 - 8y + 15 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x < y$
- d. $x \leq y$
- e. $x = y$ or a relationship cannot be established.

45. Questions

I). $x^2 + x - 2 = 0$

II). $y^2 - 10y + 24 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x < y$
- d. $x \leq y$
- e. $x = y$ or a relationship cannot be established.

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Explanations:

1. Questions

The central angle corresponding to Branch R equals 90°

The number of accounts opened by Branch R equals 450

The value of 1° equals $450 / 90 = 5$ accounts

The total number of degrees in a pie chart equals 360°

The total number of accounts opened by all branches together = $360 \times 5 = 1800$

Bank Branch P

The central angle for Branch P equals 72°

The total number of accounts opened by Branch P = $72 \times 5 = 360$

Bank Branch Q

The central angle for Branch Q equals 54°

The total number of accounts opened by Branch Q = $54 \times 5 = 270$

Bank Branch R

The central angle for Branch R equals 90°

The total number of accounts opened by Branch R = 450

Bank Branch S

The central angle for Branch S equals 108°

The total number of accounts opened by Branch S = $108 \times 5 = 540$

Bank Branch T

The central angle for Branch T equals 36°

The total number of accounts opened by Branch T = $36 \times 5 = 180$

Final Value Table:

Bank Branch	The number of accounts opened
Branch P	360
Branch Q	270
Branch R	450
Branch S	540
Branch T	180

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Answer: B

The number of accounts originally opened by Branch R = 450

The new number of accounts opened by Branch R = $450 - 90 = 360$

The number of accounts originally opened by Branch T = 180

The new number of accounts opened by Branch T = $180 + 60 = 240$

The required ratio = $360 : 240$

Dividing both terms by 120 simplifies the expression.

The required ratio = $3 : 2$

Hence, the correct answer is option B.

2. Questions

The central angle corresponding to Branch R equals 90°

The number of accounts opened by Branch R equals 450

The value of 1° equals $450 / 90 = 5$ accounts

The total number of degrees in a pie chart equals 360°

The total number of accounts opened by all branches together = $360 * 5 = 1800$

Bank Branch P

The central angle for Branch P equals 72°

The total number of accounts opened by Branch P = $72 * 5 = 360$

Bank Branch Q

The central angle for Branch Q equals 54°

The total number of accounts opened by Branch Q = $54 * 5 = 270$

Bank Branch R

The central angle for Branch R equals 90°

The total number of accounts opened by Branch R = 450

Bank Branch S

The central angle for Branch S equals 108°

The total number of accounts opened by Branch S = $108 * 5 = 540$

Bank Branch T

The central angle for Branch T equals 36°

The total number of accounts opened by Branch T = $36 * 5 = 180$

Final Value Table:

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Bank Branch	The number of accounts opened
Branch P	360
Branch Q	270
Branch R	450
Branch S	540
Branch T	180

Answer: C

The number of accounts originally opened by Branch T = 180

The new number of accounts opened by Branch T = $180 + 36 = 216$

The total number of accounts opened by Branch P = 360

The required percentage = $(216 / 360) * 100$

The required percentage = $0.6 * 100 = 60\%$

Hence, the correct answer is option C.

3. Questions

The central angle corresponding to Branch R equals 90°

The number of accounts opened by Branch R equals 450

The value of 1° equals $450 / 90 = 5$ accounts

The total number of degrees in a pie chart equals 360°

The total number of accounts opened by all branches together = $360 * 5 = 1800$

Bank Branch P

The central angle for Branch P equals 72°

The total number of accounts opened by Branch P = $72 * 5 = 360$

Bank Branch Q

The central angle for Branch Q equals 54°

The total number of accounts opened by Branch Q = $54 * 5 = 270$

Bank Branch R

The central angle for Branch R equals 90°

The total number of accounts opened by Branch R = 450

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Bank Branch S

The central angle for Branch S equals 108°

The total number of accounts opened by Branch S = $108 \times 5 = 540$

Bank Branch T

The central angle for Branch T equals 36°

The total number of accounts opened by Branch T = $36 \times 5 = 180$

Final Value Table:

Bank Branch	The number of accounts opened
Branch P	360
Branch Q	270
Branch R	450
Branch S	540
Branch T	180

Answer: A

The number of accounts opened by Branch Q = 270

The number of accounts opened by Branch T = 180

The total number of accounts opened by Branch Q and Branch T together = $270 + 180 = 450$

The number of accounts opened by Branch U = 140% of 450 = $1.4 \times 450 = 630$

Hence, the correct answer is option A.

4. Questions

The central angle corresponding to Branch R equals 90°

The number of accounts opened by Branch R equals 450

The value of 1° equals $450 / 90 = 5$ accounts

The total number of degrees in a pie chart equals 360°

The total number of accounts opened by all branches together = $360 \times 5 = 1800$

Bank Branch P

The central angle for Branch P equals 72°

The total number of accounts opened by Branch P = $72 \times 5 = 360$

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Bank Branch Q

The central angle for Branch Q equals 54°

The total number of accounts opened by Branch Q = $54 \times 5 = 270$

Bank Branch R

The central angle for Branch R equals 90°

The total number of accounts opened by Branch R = 450

Bank Branch S

The central angle for Branch S equals 108°

The total number of accounts opened by Branch S = $108 \times 5 = 540$

Bank Branch T

The central angle for Branch T equals 36°

The total number of accounts opened by Branch T = $36 \times 5 = 180$

Final Value Table:

Bank Branch	The number of accounts opened
Branch P	360
Branch Q	270
Branch R	450
Branch S	540
Branch T	180

Answer: B

The total number of accounts opened by Branch P = 360

The number of zero-balance accounts opened by Branch P = 25% of 360 = $0.25 \times 360 = 90$

The total number of accounts opened by Branch S = 540

The number of zero-balance accounts opened by Branch S = 20% of 540 = $0.2 \times 540 = 108$

The total number of zero-balance accounts opened by Branch P and Branch S together = $90 + 108 = 198$

Hence, the correct answer is option B.

5. Questions

The central angle corresponding to Branch R equals 90°

The number of accounts opened by Branch R equals 450

The value of 1° equals $450 / 90 = 5$ accounts

The total number of degrees in a pie chart equals 360°

The total number of accounts opened by all branches together = $360 * 5 = 1800$

Bank Branch P

The central angle for Branch P equals 72°

The total number of accounts opened by Branch P = $72 * 5 = 360$

Bank Branch Q

The central angle for Branch Q equals 54°

The total number of accounts opened by Branch Q = $54 * 5 = 270$

Bank Branch R

The central angle for Branch R equals 90°

The total number of accounts opened by Branch R = 450

Bank Branch S

The central angle for Branch S equals 108°

The total number of accounts opened by Branch S = $108 * 5 = 540$

Bank Branch T

The central angle for Branch T equals 36°

The total number of accounts opened by Branch T = $36 * 5 = 180$

Final Value Table:

Bank Branch	The number of accounts opened
Branch P	360
Branch Q	270
Branch R	450
Branch S	540
Branch T	180

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Answer: E

The total number of accounts opened by Branch S = 540

The total number of accounts opened by Branch Q = 270

The required difference = $540 - 270 = 270$

Hence, the correct answer is option E.

6. Questions

Year 2021:

The total number of magazines distributed in 2021 = 320

The difference between the number of magazines and journals distributed in 2021 = 140

The number of journals distributed in 2021 = $320 - 140 = 180$

Year 2022:

The total number of magazines distributed in 2022 = 280

The difference between the number of magazines and journals distributed in 2022 = 120

The number of journals distributed in 2022 = $280 - 120 = 160$

Year 2023:

The total number of magazines distributed in 2023 = 440

The difference between the number of magazines and journals distributed in 2023 = 200

The number of journals distributed in 2023 = $440 - 200 = 240$

Year 2024:

The total number of magazines distributed in 2024 = 360

The difference between the number of magazines and journals distributed in 2024 = 160

The number of journals distributed in 2024 = $360 - 160 = 200$

Year 2025:

The total number of magazines distributed in 2025 = 400

The difference between the number of magazines and journals distributed in 2025 = 180

The number of journals distributed in 2025 = $400 - 180 = 220$

Final Value Table:

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Year	The total number of magazines distributed	The total number of journals distributed
2021	320	180
2022	280	160
2023	440	240
2024	360	200
2025	400	220

Answer: B

The total number of magazines distributed in 2022 = 280

The total number of journals distributed in 2024 = 200

The required difference = $280 - 200 = 80$

Hence, the correct answer is option B.

7. Questions

Year 2021:

The total number of magazines distributed in 2021 = 320

The difference between the number of magazines and journals distributed in 2021 = 140

The number of journals distributed in 2021 = $320 - 140 = 180$

Year 2022:

The total number of magazines distributed in 2022 = 280

The difference between the number of magazines and journals distributed in 2022 = 120

The number of journals distributed in 2022 = $280 - 120 = 160$

Year 2023:

The total number of magazines distributed in 2023 = 440

The difference between the number of magazines and journals distributed in 2023 = 200

The number of journals distributed in 2023 = $440 - 200 = 240$

Year 2024:

The total number of magazines distributed in 2024 = 360

The difference between the number of magazines and journals distributed in 2024 = 160

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The number of journals distributed in 2024 = $360 - 160 = 200$

Year 2025:

The total number of magazines distributed in 2025 = 400

The difference between the number of magazines and journals distributed in 2025 = 180

The number of journals distributed in 2025 = $400 - 180 = 220$

Final Value Table:

Year	The total number of magazines distributed	The total number of journals distributed
2021	320	180
2022	280	160
2023	440	240
2024	360	200
2025	400	220

Answer: E

The total number of magazines distributed in 2022 = 280

The number of academic magazines distributed in 2022 = $(1 / 4) * 280 = 70$

The number of sports magazines distributed in 2022 = $280 - 70 = 210$

Hence, the correct answer is option E.

8. Questions

Year 2021:

The total number of magazines distributed in 2021 = 320

The difference between the number of magazines and journals distributed in 2021 = 140

The number of journals distributed in 2021 = $320 - 140 = 180$

Year 2022:

The total number of magazines distributed in 2022 = 280

The difference between the number of magazines and journals distributed in 2022 = 120

The number of journals distributed in 2022 = $280 - 120 = 160$

Year 2023:

The total number of magazines distributed in 2023 = 440

The difference between the number of magazines and journals distributed in 2023 = 200

The number of journals distributed in 2023 = $440 - 200 = 240$

Year 2024:

The total number of magazines distributed in 2024 = 360

The difference between the number of magazines and journals distributed in 2024 = 160

The number of journals distributed in 2024 = $360 - 160 = 200$

Year 2025:

The total number of magazines distributed in 2025 = 400

The difference between the number of magazines and journals distributed in 2025 = 180

The number of journals distributed in 2025 = $400 - 180 = 220$

Final Value Table:

Year	The total number of magazines distributed	The total number of journals distributed
2021	320	180
2022	280	160
2023	440	240
2024	360	200
2025	400	220

Answer: A

The total number of magazines distributed in 2024 = 360

The total number of journals distributed in 2024 = 200

The sum of magazines and journals distributed in 2024 = $360 + 200 = 560$

The total number of magazines distributed in 2021 = 320

Difference = $560 - 320 = 240$

Required percentage = $(240 / 320) * 100 = (3 / 4) * 100 = 75\%$

Hence, the correct answer is option A.

9. Questions

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Year 2021:

The total number of magazines distributed in 2021 = 320

The difference between the number of magazines and journals distributed in 2021 = 140

The number of journals distributed in 2021 = $320 - 140 = 180$

Year 2022:

The total number of magazines distributed in 2022 = 280

The difference between the number of magazines and journals distributed in 2022 = 120

The number of journals distributed in 2022 = $280 - 120 = 160$

Year 2023:

The total number of magazines distributed in 2023 = 440

The difference between the number of magazines and journals distributed in 2023 = 200

The number of journals distributed in 2023 = $440 - 200 = 240$

Year 2024:

The total number of magazines distributed in 2024 = 360

The difference between the number of magazines and journals distributed in 2024 = 160

The number of journals distributed in 2024 = $360 - 160 = 200$

Year 2025:

The total number of magazines distributed in 2025 = 400

The difference between the number of magazines and journals distributed in 2025 = 180

The number of journals distributed in 2025 = $400 - 180 = 220$

Final Value Table:

Year	The total number of magazines distributed	The total number of journals distributed
2021	320	180
2022	280	160
2023	440	240
2024	360	200
2025	400	220

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Answer: C

The total number of journals distributed in 2021 = 180

The total number of journals distributed in 2024 = 200

The total number of journals distributed in 2025 = 220

Sum of journals distributed in the given three years = $180 + 200 + 220 = 600$

Required average = $600 / 3 = 200$

Hence, the correct answer is option C.

10. Questions

Year 2021:

The total number of magazines distributed in 2021 = 320

The difference between the number of magazines and journals distributed in 2021 = 140

The number of journals distributed in 2021 = $320 - 140 = 180$

Year 2022:

The total number of magazines distributed in 2022 = 280

The difference between the number of magazines and journals distributed in 2022 = 120

The number of journals distributed in 2022 = $280 - 120 = 160$

Year 2023:

The total number of magazines distributed in 2023 = 440

The difference between the number of magazines and journals distributed in 2023 = 200

The number of journals distributed in 2023 = $440 - 200 = 240$

Year 2024:

The total number of magazines distributed in 2024 = 360

The difference between the number of magazines and journals distributed in 2024 = 160

The number of journals distributed in 2024 = $360 - 160 = 200$

Year 2025:

The total number of magazines distributed in 2025 = 400

The difference between the number of magazines and journals distributed in 2025 = 180

The number of journals distributed in 2025 = $400 - 180 = 220$

Final Value Table:

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Year	The total number of magazines distributed	The total number of journals distributed
2021	320	180
2022	280	160
2023	440	240
2024	360	200
2025	400	220

Answer: A

The total number of magazines distributed in 2022 = 280

The total number of magazines distributed in 2020 = $280 \times (120 / 100) = 336$

Hence, the correct answer is option A.

11. Questions

Answer: C

Downstream speed = $60 / 4 = 15$ km/hr

Let the speed of the stream be x km/hr, then the speed of the boat in still water is $4x$ km/hr.

Downstream speed = $4x + x = 5x$

So, $5x = 15$ (which means $x = 3$)

Upstream speed = $4x - x = 3x = 3 \times 3 = 9$ km/hr

Time taken to cover 36 km upstream = $36 / 9 = 4$ hours.

Hence, the correct answer is option C.

12. Questions

Answer: C

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Let the age of Dev and Divya 4 years ago be $3x$ years and $4x$ years respectively

Present age of Dev = $3x + 4$

Present age of Divya = $4x + 4$

Age of Dev after 8 years from now = $3x + 4 + 8 = 3x + 12$

Age of Divya after 8 years from now = $4x + 4 + 8 = 4x + 12$

According to the given condition, the ratio of their ages after 8 years is 5:6

$$(3x + 12) / (4x + 12) = 5 / 6$$

$$6 * (3x + 12) = 5 * (4x + 12)$$

$$18x + 72 = 20x + 60$$

$$20x - 18x = 72 - 60$$

$$2x = 12$$

$$x = 12 / 2 = 6$$

$$\text{Present age of Dev} = 3 * 6 + 4 = 18 + 4 = 22 \text{ years}$$

Hence, the correct answer is option C.

13. Questions

Answer: C

Let the cost price of the watch for A be x.

$$\text{Cost price for B} = x * 1.20 = 1.2x$$

$$\text{Cost price for C} = 1.2x * 0.90 = 1.08x$$

$$\text{According to the question, } 1.08x = 1080$$

$$x = 1000$$

Hence, the correct answer is option C.

14. Questions

Answer: C

$$\text{Cost price of variety 'A' rice} = \text{Rs. } 80/\text{kg}$$

$$\text{Cost price of variety 'B' rice} = \text{Rs. } 50/\text{kg}$$

$$\text{Ratio of variety 'A' to variety 'B'} = 3:2$$

$$\text{Let the cost price of the resulting mixture be Rs. } x \text{ per kg}$$

By the rule of alligation:

$$80 \quad 50$$

$$x$$

$$(x - 50) \quad (80 - x)$$

Let

$$3 / 2 = (x - 50) / (80 - x)$$

$$3 * (80 - x) = 2 * (x - 50)$$

$$240 - 3x = 2x - 100$$

$$5x = 340$$

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$$x = 340 / 5 = 68$$

Hence, the correct answer is option C.

15. Questions

Answer: E

Let the initial investment of P be Rs. y.

$$\text{Ratio of profit share of P and Q} = [y * 6 + (y + 3000) * 6] : [15000 * 6]$$

$$\text{Profit ratio} = (12y + 18000) : 90000$$

Given profit ratio is 4 : 3

$$(12y + 18000) / 90000 = 4 / 3$$

$$12y + 18000 = 120000$$

$$12y = 102000$$

$$Y = 8500$$

Initial investment of P = Rs. 8500

Hence, the correct answer is option E.

16. Questions

Answer: B

Let the side of the square be a cm.

$$\text{Total area of the square} = a^2$$

$$a^2 = 900$$

$$a = 30$$

$$\text{Length of the rectangle} = 30/5 * 6 = 36 \text{ cm}$$

$$\text{Perimeter of the rectangle} = 2 * (L + B) = 2 * (36 + 15) = 2 * 51 = 102 \text{ cm}$$

Hence, the correct answer is option B.

17. Questions

Answer: C

$$\text{Simple interest} = (\text{Principal} * \text{Rate} * \text{Time}) / 100$$

$$5400 = (450x * 15 * 4) / 100$$

$$5400 = (450x * 60) / 100$$

$$5400 = 270x$$

$$x = 20$$

The numbers for the LCM calculation are $(20 + 2) = 22$ and $(20 + 6) = 26$

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LCM of 22 and 26 = 286

Hence, the correct answer is option C.

18. Questions

Answer: B

Let the total monthly income of Rohit be $100x$.

Amount spent on rent and travel = $15x + 10x = 25x$

Remaining income = $100x - 25x = 75x$

Amount spent on food = 20% of $75x = 15x$

Remaining income = $75x - 15x = 60x$

Amount spent on other expenses = $60x * (1 / 2) = 30x$

Savings = $60x - 30x = 30x$

According to the question, $30x = 9000$

Value of $x = 300$

Amount spent on travel = $10 * 300 = \text{Rs. } 3000$

Hence, the correct answer is option B.

19. Questions

Answer: D

Speed of the car = 45 km/hr

Time taken = 3 hours 20 minutes = $3 + (20 / 60)$ hours = $10/3$ hours

Total distance covered = Speed * Time

Total distance covered = $45 * (10 / 3) = 150$ km

Hence, the correct answer is option D.

20. Questions

Answer: A

Let the three consecutive even numbers be $(x - 2)$, x and $(x + 2)$ respectively

Average of the three numbers = $(x - 2 + x + x + 2) / 3 = 3x / 3 = x$

Total sum of the three numbers = $3x$

According to the given condition, $3x = x + 56$

The difference of $3x$ and $x = 56$

$2x = 56$

Value of $x = 28$

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The value of the smallest number = $28 - 2 = 26$

The value of the largest number = $28 + 2 = 30$

The total product of the smallest and largest numbers = $26 * 30 = 780$

Hence, the correct answer is option A.

21. Questions

Answer: B

$$\Rightarrow (18 * 12) + (15 * 16) - (14 * 10) = ?$$

$$\Rightarrow 216 + 240 - 140 = ?$$

$$\Rightarrow 456 - 140 = ?$$

$$\Rightarrow ? = 316$$

Hence, the correct answer is option B.

22. Questions

Answer: C

$$\Rightarrow \sqrt{784} + \sqrt{169} + \sqrt{256} = ? + 18$$

$$\Rightarrow 28 + 13 + 16 = ? + 18$$

$$\Rightarrow 57 = ? + 18$$

$$\Rightarrow ? = 57 - 18$$

$$\Rightarrow ? = 39$$

Hence, the correct answer is option C.

23. Questions

Answer: E

$$\Rightarrow 45\% * 600 + 30\% * 450 = ? + 105$$

$$\Rightarrow (45 / 100) * 600 + (30 / 100) * 450 = ? + 105$$

$$\Rightarrow (45 * 6) + (3 * 45) = ? + 105$$

$$\Rightarrow 270 + 135 = ? + 105$$

$$\Rightarrow 405 = ? + 105$$

$$\Rightarrow ? = 405 - 105$$

$$\Rightarrow ? = 300$$

Hence, the correct answer is option E.

24. Questions

Answer: B

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$$\Rightarrow (5/8) * 720 - (4/9) * 630 = ? * 5$$

$$\Rightarrow (5 * 90) - (4 * 70) = ? * 5$$

$$\Rightarrow 450 - 280 = ? * 5$$

$$\Rightarrow 170 = ? * 5$$

$$\Rightarrow ? = 170 / 5$$

$$\Rightarrow ? = 34$$

Hence, the correct answer is option B.

25. Questions

Answer: A

$$\Rightarrow 16^2 - 14^2 + (180 / 4) = ? + 5^2$$

$$\Rightarrow 256 - 196 + 45 = ? + 25$$

$$\Rightarrow 60 + 45 = ? + 25$$

$$\Rightarrow 105 = ? + 25$$

$$\Rightarrow ? = 105 - 25$$

$$\Rightarrow ? = 80$$

Hence, the correct answer is option A.

26. Questions

Answer: C

$$\Rightarrow \sqrt[3]{1330.98 + 24.99\% \text{ of } 480.05} = ? / 4.01$$

$$\Rightarrow \sqrt[3]{1331 + 25\% \text{ of } 480} = ? / 4$$

$$\Rightarrow 11 + (25 / 100) * 480 = ? / 4$$

$$\Rightarrow 11 + 120 = ? / 4$$

$$\Rightarrow 131 = ? / 4$$

$$\Rightarrow ? = 131 * 4$$

$$\Rightarrow ? = 524$$

Hence, the correct answer is option C.

27. Questions

Answer: D

$$\Rightarrow (3.99)^4 * (16.01)^2 \div (63.99)^2 = 4^?$$

$$\Rightarrow 4^4 * 16^2 \div 64^2 = 4^?$$

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$$\Rightarrow 4^4 * (4^2)^2 \div (4^3)^2 = 4^?$$

$$\Rightarrow 4^4 * 4^4 \div 4^6 = 4^?$$

$$\Rightarrow 4^{(4+4-6)} = 4^?$$

$$\Rightarrow 4^2 = 4^?$$

$$\Rightarrow ? = 2$$

Hence, the correct answer is option D.

28. Questions

Answer: B

$$\Rightarrow 1199.98 \div 23.99 + 44.98 * 12.01 = ?$$

$$\Rightarrow 1200 \div 24 + 45 * 12 = ?$$

$$\Rightarrow 50 + 540 = ?$$

$$\Rightarrow ? = 590$$

Hence, the correct answer is option B.

29. Questions

Answer: C

$$\Rightarrow \sqrt{(? + 84.01)} = 15.99 * 4.01 - 25.01 * 1.99$$

$$\Rightarrow \sqrt{(? + 84)} = 16 * 4 - 25 * 2$$

$$\Rightarrow \sqrt{(? + 84)} = 64 - 50$$

$$\Rightarrow \sqrt{(? + 84)} = 14$$

$$\Rightarrow ? + 84 = 196$$

$$\Rightarrow ? = 196 - 84$$

$$\Rightarrow ? = 112$$

Hence, the correct answer is option C.

30. Questions

Answer: B

$$\Rightarrow (74.99\% \text{ of } 640.01) \div [(1/3) * 179.98] = ?$$

$$\Rightarrow (75\% \text{ of } 640) \div [(1/3) * 180] = ?$$

$$\Rightarrow ((3/4) * 640) \div [(1/3) * 180] = ?$$

$$\Rightarrow (3 * 160) \div (60) = ?$$

$$\Rightarrow 480 \div 60 = ?$$

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$$\Rightarrow ? = 8$$

Hence, the correct answer is option B.

31. Questions

Answer: A

The series follows the pattern of multiplying by 2 continuously:

$$11 \times 2 = 22$$

$$22 \times 2 = 44$$

$$44 \times 2 = 88$$

$$88 \times 2 = 176$$

$$176 \times 2 = 352$$

$$\text{So, } ? = 88.$$

Hence, the correct answer is option A.

32. Questions

Answer: E

The series follows an alternating pattern of adding and subtracting squares of decreasing consecutive numbers:

$$47 + 9^2 = 128$$

$$128 - 8^2 = 64$$

$$64 + 7^2 = 113$$

$$113 - 6^2 = 77$$

$$77 + 5^2 = 102$$

$$\text{So, } ? = 128.$$

Hence, the correct answer is option E.

33. Questions

Answer: D

The series follows the pattern of multiplying by increasing multiples of 0.5:

$$16 \times 0.5 = 8$$

$$8 \times 1 = 8$$

$$8 \times 1.5 = 12$$

$$12 \times 2 = 24$$

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$$24 \times 2.5 = 60$$

So, ? = 16.

Hence, the correct answer is option D.

34. Questions

Answer: C

The series follows the pattern of subtracting consecutive even numbers starting from 16:

$$212 - 16 = 196$$

$$196 - 18 = 178$$

$$178 - 20 = 158$$

$$158 - 22 = 136$$

$$136 - 24 = 112$$

So, ? = 136.

Hence, the correct answer is option C.

35. Questions

Answer: B

The series follows a double difference pattern (second difference = +5):

$$50 \quad 55 \quad 65 \quad 80 \quad 100 \quad 125$$

$$+5 \quad +10 \quad +15 \quad +20 \quad +25$$

So, ? = 125.

Hence, the correct answer is option B.

36. Questions

Answer: B

$$15 + 8 = 23 \text{ (Therefore, 25 is wrong and should be 23)}$$

$$23 + 16 = 39$$

$$39 + 24 = 63$$

$$63 + 32 = 95$$

$$95 + 40 = 135$$

Pattern: The difference follows consecutive multiples of 8 (8, 16, 24, 32, 40).

Hence, the correct answer is option B.

37. Questions

Answer: D

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$$10 + 2^2 = 14$$

$$14 + 4^2 = 30$$

$$30 + 6^2 = \mathbf{66}$$
 (Therefore, 64 is wrong and should be 66)

$$66 + 8^2 = 130$$

$$130 + 10^2 = 230$$

Pattern: The difference follows squares of consecutive even numbers ($2^2, 4^2, 6^2, 8^2, 10^2$).

Hence, the correct answer is option D.

38. Questions

Answer: A

$$100 - 1^3 = 99$$

$$99 - 2^3 = \mathbf{91}$$
 (Therefore, 90 is wrong and should be 91)

$$91 - 3^3 = 64$$

$$64 - 4^3 = 0$$

$$0 - 5^3 = -125$$

Pattern: The difference follows cubes of consecutive numbers ($1^3, 2^3, 3^3, 4^3, 5^3$).

Hence, the correct answer is option A.

39. Questions

Answer: C

$$5 \times 3 - 2 = 13$$

$$13 \times 3 - 2 = 37$$

$$37 \times 3 - 2 = 109$$

$$109 \times 3 - 2 = \mathbf{325}$$
 (Therefore, 320 is wrong and should be 325) @RankZen Banker

$$325 \times 3 - 2 = 973$$

Pattern: The terms are multiplied by 3 and then decreased by 2 continuously.

Hence, the correct answer is option C.

40. Questions

Answer: E

$$50 + 17 = 67$$

$$67 + 19 = 86$$

$$86 + 23 = 109$$

$$109 + 29 = 138$$

$$138 + 31 = \mathbf{169}$$
 (Therefore, 170 is wrong and should be 169)

Pattern: The difference follows consecutive prime numbers starting from 17 (17, 19, 23, 29, 31).

Hence, the correct answer is option E.

41. Questions

Answer: D

I). $x^2 + 8x + 15 = 0$

$$x^2 + 5x + 3x + 15 = 0$$

$$x(x + 5) + 3(x + 5) = 0$$

$$(x+3)(x+5) = 0$$

$$x = -5, -3$$

II). $y^2 - 6y - 27 = 0$

$$y^2 - 9y + 3y - 27 = 0$$

$$y(y - 9) + 3(y - 9) = 0$$

$$(y-9)(y+3) = 0$$

$$y = -3, 9$$

Comparison:

$$-5 < -3$$

$$-5 < 9$$

$$-3 = -3$$

$$-3 < 9$$

$$x \leq y$$

Hence, the correct answer is option D.

42. Questions

Answer: E

I). $x^2 + 19x + 84 = 0$

$$x^2 + 12x + 7x + 84 = 0$$

$$x(x + 12) + 7(x + 12) = 0$$

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$$(x+7)(x+12) = 0$$

$$x = -12, -7$$

$$\text{II). } y^2 + 15y + 56 = 0$$

$$y^2 + 8y + 7y + 56 = 0$$

$$y(y + 8) + 7(y + 8) = 0$$

$$(y+8)(y+7) = 0$$

$$y = -8, -7$$

Comparison:

$$-12 < -8$$

$$-12 < -7$$

$$-7 > -8$$

$$-7 = -7$$

A relationship cannot be established.

Hence, the correct answer is option E.

43. Questions

Answer: A

$$\text{I). } x^2 - 14x + 48 = 0$$

$$x^2 - 8x - 6x + 48 = 0$$

$$x(x - 8) - 6(x - 8) = 0$$

$$(x-6)(x-8) = 0$$

$$x = 6, 8$$

$$\text{II). } y^2 - 6y + 8 = 0$$

$$y^2 - 4y - 2y + 8 = 0$$

$$y(y - 4) - 2(y - 4) = 0$$

$$(y-4)(y-2) = 0$$

$$y = 2, 4$$

Comparison:

$$6 > 2$$

$$6 > 4$$

$$8 > 2$$

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$$8 > 4$$

$$x > y$$

Hence, the correct answer is option A.

44. Questions

Answer: B

I). $x^2 - 12x + 35 = 0$

$$x^2 - 7x - 5x + 35 = 0$$

$$x(x - 7) - 5(x - 7) = 0$$

$$(x-5)(x-7) = 0$$

$$x = 5, 7$$

II). $y^2 - 8y + 15 = 0$

$$y^2 - 5y - 3y + 15 = 0$$

$$y(y - 5) - 3(y - 5) = 0$$

$$(y-3)(y-5) = 0$$

$$y = 3, 5$$

Comparison:

$$5 > 3$$

$$5 = 5$$

$$7 > 3$$

$$7 > 5$$

$$x \geq y$$

Hence, the correct answer is option B.

45. Questions

Answer: C

I). $x^2 + x - 2 = 0$

$$x^2 + 2x - x - 2 = 0$$

$$x(x + 2) - 1(x + 2) = 0$$

$$(x+2)(x-1) = 0$$

$$x = -2, 1$$

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II). $y^2 - 10y + 24 = 0$

$$y^2 - 6y - 4y + 24 = 0$$

$$y(y - 6) - 4(y - 6) = 0$$

$$(y-6)(y-4) = 0$$

$$y = 4, 6$$

Comparison:

$$-2 < 4$$

$$-2 < 6$$

$$1 < 4$$

$$1 < 6$$

$$x < y$$

Hence, the correct answer is option C.

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